

Generation Interconnection System Impact Study Report

For Queue Project Q20-1387

Anthem 138 kV
December 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1387. The Interconnection Customer proposes a 82.9 MW solar photovoltaic facility interconnecting at the Anthem 138 kV substation in the Virginia area. The request was received on January 15, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 532 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,547,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Shepard - Bloomington 138kV Cir. 1
Contingency: Shepard - Bloomington 138kV #2
Loading: 130.2% (Rating: 332 MVA)
DFAX: 20.1%
Pre-project loading: 125.1%
2. Facility: Bloomington Substation - Shepard 138kV Ckt 2
Contingency: Bloomington - Sheprd 138kV Line 1
Loading: 127.9% (Rating: 337 MVA)
DFAX: 20.1%
Pre-project loading: 122.9%
3. Facility: Tokoma - Yulee 138kV
Contingency: Darien - Colesville 138kV
Loading: 120.9% (Rating: 409 MVA)
DFAX: 6.8%
Pre-project loading: 119.5%
4. Facility: Swinks Mill - Shepard 138kV Line
Contingency: Swinks Mill - Shepard 138kV
Loading: 117.1% (Rating: 389 MVA)
DFAX: 20.2%
Pre-project loading: 112.8%
5. Facility: Swinks Mill - Carderock 138kV
Contingency: Swinks Mill - Carderock 138kV line
Loading: 117.0% (Rating: 398 MVA)
DFAX: 19.5%
Pre-project loading: 112.9%
6. Facility: Shepard - Swinks Mill 138kV
Contingency: Shepard Substation - Swinks Mill 138kV Line 2
Loading: 116.2% (Rating: 392 MVA)

DFAX: 20.2%

Pre-project loading: 112.0%

7. Facility: Lakehead Pipeline - Carderock 138kV Line 1
Contingency: Carderock - Lakehead Pipeline Substation 138kV
Loading: 113.8% (Rating: 268 MVA)
DFAX: 8.0%
Pre-project loading: 111.3%
8. Facility: Cowlitz Falls - Saranac Facility Substation 138kV line
Contingency: Cia - New Sterling 138kV
Loading: 113.7% (Rating: 221 MVA)
DFAX: 5.8%
Pre-project loading: 111.5%
9. Facility: Carderock - Swinks Mill 138kV L
Contingency: Swinks Mill - Carderock 138kV Line
Loading: 112.8% (Rating: 413 MVA)
DFAX: 19.5%
Pre-project loading: 108.8%
10. Facility: Carderock - Lakehead Pipeline 138kV Circuit 2
Contingency: Lakehead Pipeline - Carderock 138kV Ckt 1
Loading: 112.3% (Rating: 271 MVA)
DFAX: 8.0%
Pre-project loading: 109.8%
11. Facility: Bloomington - Mottman 138kV #2
Contingency: Mottman - Bloomington 138kV #1
Loading: 111.1% (Rating: 476 MVA)
DFAX: 16.7%
Pre-project loading: 108.2%
12. Facility: Cia - New Sterling 138kV Line
Contingency: Cowlitz Falls - Cia 138kV
Loading: 108.0% (Rating: 408 MVA)
DFAX: 9.5%
Pre-project loading: 106.0%
13. Facility: Helijet - Elk Garden 138kV
Contingency: Carderock Substation - Elk Garden 138kV Line
Loading: 106.9% (Rating: 282 MVA)
DFAX: 28.1%
Pre-project loading: 98.7%
14. Facility: Helijet - Lakehead Pipeline 138kV #1
Contingency: Helijet - Lakehead Pipeline 138kV Cir. 2
Loading: 106.8% (Rating: 307 MVA)
DFAX: 8.2%
Pre-project loading: 104.6%
15. Facility: Cia - Papago Buttes 138kV
Contingency: Cia - Papago Buttes 138kV Line 2
Loading: 105.6% (Rating: 450 MVA)
DFAX: 9.1%
Pre-project loading: 103.9%
16. Facility: Cia - Papago Buttes 138kV line
Contingency: Cia - Papago Buttes 138kV Cir. 1
Loading: 105.6% (Rating: 450 MVA)
DFAX: 9.1%

Pre-project loading: 103.9%

17. Facility: Elk Garden - Anthem 138kV

Contingency: Foster Dam - McCormick 138kV

Loading: 105.5% (Rating: 273 MVA)

DFAX: 43.4%

Pre-project loading: 92.3%

18. Facility: Cowlitz Falls - Cia 138kV

Contingency: Cia - New Sterling 138kV

Loading: 105.5% (Rating: 308 MVA)

DFAX: 5.8%

Pre-project loading: 104.0%

19. Facility: Bloomington - Mottman 138kV Circuit 1

Contingency: Mottman - Bloomington 138kV

Loading: 105.1% (Rating: 503 MVA)

DFAX: 16.7%

Pre-project loading: 102.4%

20. Facility: Mottman - Papago Buttes Substation 138kV Circuit 1

Contingency: Mottman - Papago Buttes 138kV Cir. 2

Loading: 104.7% (Rating: 496 MVA)

DFAX: 9.5%

Pre-project loading: 103.1%

21. Facility: Papago Buttes - Mottmn 138kV Ckt 2

Contingency: Papago Buttes - Mottman 138kV Cir. 1

Loading: 104.7% (Rating: 496 MVA)

DFAX: 9.5%

Pre-project loading: 103.1%

22. Facility: Carderock - Elk Garden 138kV

Contingency: Helijet - Elk Garden 138kV

Loading: 100.6% (Rating: 291 MVA)

DFAX: 23.4%

Pre-project loading: 93.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shepard - Bloomington 138kV Cir. 1	Rebuild line with higher-capacity conductor	\$1,633,000	49% (\$805,000)
Shepard Substation - Bloomington 138kV #2	Rebuild line with higher-capacity conductor	\$1,415,000	49% (\$698,000)
Tokoma - Yulee 138kV	Rebuild line with higher-capacity conductor	\$1,266,000	6% (\$79,000)
Swinks Mill - Shepard 138kV Line	Rebuild line with higher-capacity conductor	\$731,000	49% (\$361,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Swinks Mil - Carderock 138kV Ckt 2	Rebuild line with higher-capacity conductor	\$3,827,000	58% (\$2,220,000)
Swinks Mill - Shepard 138kV Line 1	Rebuild line with higher-capacity conductor	\$889,000	49% (\$439,000)
Lakehead Pipeline - Carderock 138kV line	Rebuild line with higher-capacity conductor	\$2,704,000	48% (\$1,305,000)
Cowlitz Falls - Saranac Facility 138kV line	Rebuild line with higher-capacity conductor	\$655,000	23% (\$150,000)
Carderock - Swinks Mill 138kV L	Rebuild line with higher-capacity conductor	\$1,577,000	58% (\$915,000)
Carderock - Lakehead Pipeline 138kV Ckt 2	Rebuild line with higher-capacity conductor	\$3,018,000	48% (\$1,456,000)
Bloomington - Mottman 138kV #2	Rebuild line with higher-capacity conductor	\$1,677,000	29% (\$485,000)
Cia - New Sterling 138kV Line	Reconductor line	\$884,000	23% (\$204,000)
Helijet - Elk Garden 138kV	Reconductor line	\$790,000	77% (\$608,000)
Helijet - Lakehead Pipeline 138kV #1	Reconductor line	\$557,000	48% (\$269,000)
Cia - Papago Buttes 138kV	Reconductor line	\$898,000	23% (\$207,000)
Cia - Papago Buttes 138kV #2	Reconductor line	\$830,000	23% (\$191,000)
Elk Garden - Anthem 138kV	Reconductor line	\$1,687,000	100% (\$1,687,000)
Cia - Cowlitz Falls 138kV	Reconductor line	\$817,000	23% (\$187,000)
Bloomington - Mottman 138kV Circuit 1	Reconductor line	\$985,000	29% (\$285,000)
Papago Buttes - Mottman Substation 138kV Cir. 1	Reconductor line	\$484,000	23% (\$111,000)
Mottman - Papago Buttes 138kV	Reconductor line	\$472,000	23% (\$109,000)
Carderock - Elk Garden 138kV	Reconductor line	\$1,048,000	74% (\$776,000)
	Total allocated to project		\$13,547,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.